Cognitive Achievement Distributions and Adult Wage Returns: Evidence from NELS and ELS

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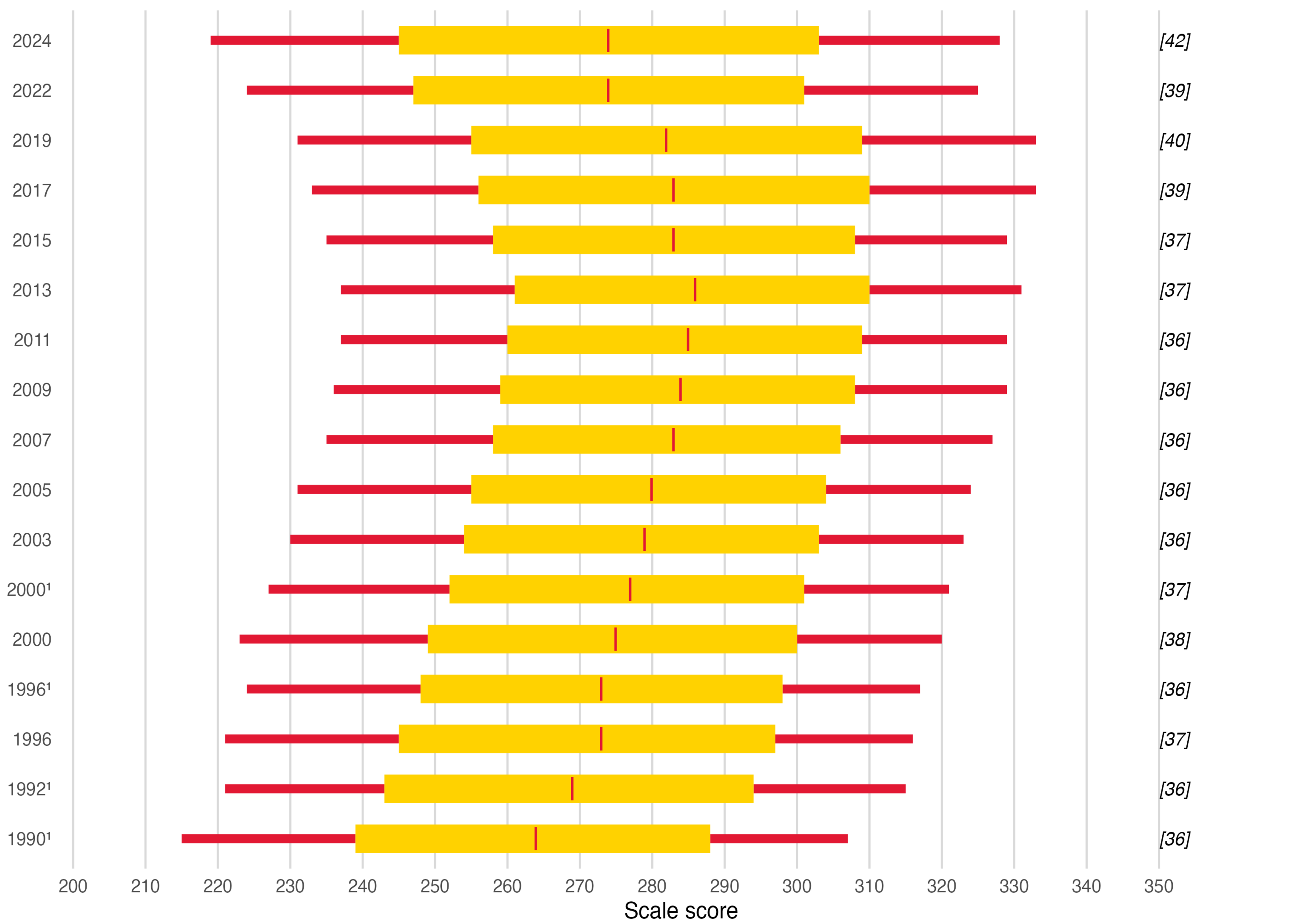
Background: Changing U.S. Student Test Scores Over Time

Trends in Grade 8 Mathematics Performance from 1990-2024
10th, 25th, 50th, 75th, and 90th percentile scale scores (Standard Deviation in brackets)

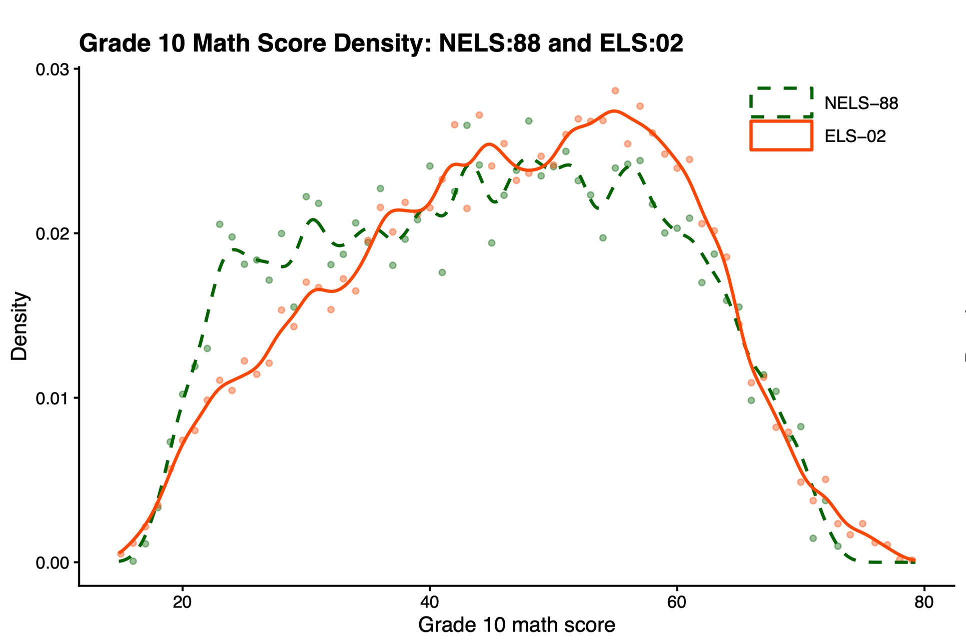


¹ Accommodations were not permitted for this assessment.
NOTE: Some apparent differences between estimates may not be statistically significant.
SOURCE: U.S. Department of Education, Institute of Education Sciences, National Center for Education Statistics, National Assessment of Educational Progress (NAEP), 1990-2024 Mathematics Assessments.

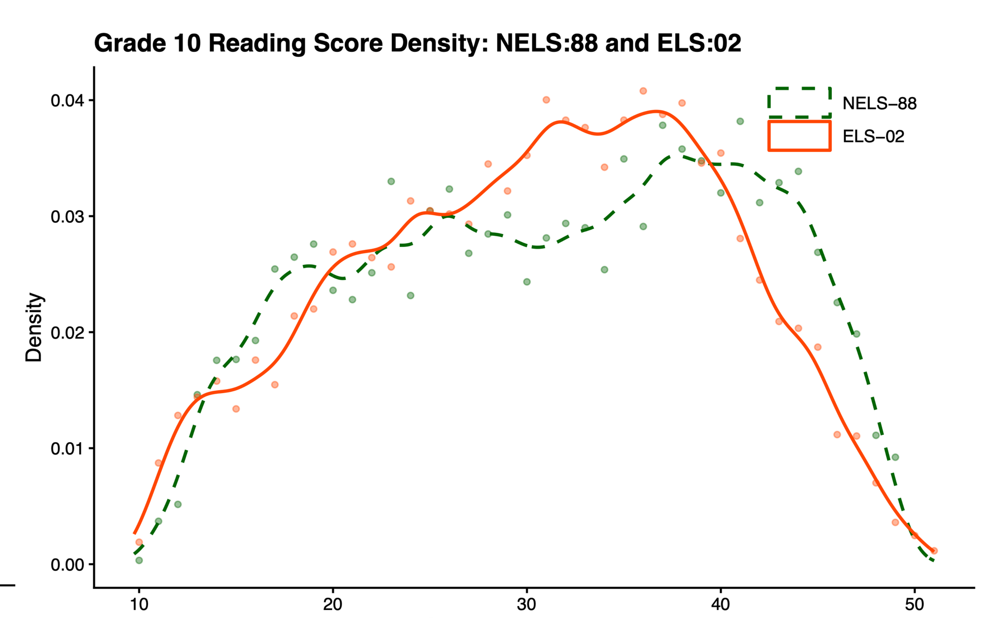
- The National Assessment of Educational Progress (NAEP) assesses a nationally representative sample of U.S. students.[1]
- National test scores increased during the 1990s and 2000s for all students.
- Decline in the bottom 25% began around 2015; increased standard deviation and left-skewness
- Steep declines following the 2020 COVID pandemic.

NELS and ELS Test Score Distributions

- Data drawn from the National Education Longitudinal Study[2] (NELS:88) and the Education Longitudinal Study[3] (ELS:2002); two cohorts: 10th graders in 1990 and 10th graders in 2002.
- NELS/ELS offer individual-level specifics, including adult wage data (7-year follow-up).
- For both math and reading, distributions changed negligibly across cohorts.
- This is also true for demographic subgroups (race, gender).



NELS–88: Mean = 44.4, SD = 13.6, Skewness = –0.03, p10 = 25, p25 = 33, p50 = 44, p75 = 55, p90 = 62
ELS–02: Mean = 46.4, SD = 13.2, Skewness = –0.16, p10 = 27, p25 = 36, p50 = 47, p75 = 56, p90 = 63



NELS–88: Mean = 31.3, SD = 9.8, Skewness = –0.15, p10 = 17, p25 = 23, p50 = 32, p75 = 39, p90 = 44
ELS–02: Mean = 30.4, SD = 9.2, Skewness = –0.17, p10 = 17, p25 = 23, p50 = 31, p75 = 37, p90 = 42

Research Questions

RQ1: How have the distributions of cognitive achievement as measured by student test scores changed over time?
RQ2: How have wage returns to cognitive achievement changed over time, as measured by the changing link between test scores and adult wages?
RQ3: What is the relationship between changing distributions of cognitive achievement and changing wage returns?

Wage Returns to Cognitive Achievement

$$\ln(\text{Wage})_{i,t,g} = \alpha_{t,g} + \beta_{t,g} * \text{Score}_{i,t,g} + \varepsilon_{i,t,g}$$

where i = individual, t = cohort, and g = men/women

Wage Returns to Grade 10 Math Scores by Men/Women Separately, NELS:88 and ELS:02 (Bracketed SE)

Term	Men: NELS	Men: ELS	Women: NELS	Women: ELS
Math ($\hat{\beta}$)	0.006*** [0.001]	0.007*** [0.001]	0.009*** [0.002]	0.014*** [0.001]
Change from NELS		0.001 [0.001]		0.004** [0.002]
IQR	23.220	20.890	21.750	19.600
Math*IQR	0.137	0.152	0.200	0.268
N	4065	4601	4170	5108

Significance levels: * p < .10, ** p < .05, *** p < .01

Wage Returns to Grade 10 Reading Scores by Men/Women Separately, NELS:88 and ELS:02 (Bracketed SE)

Term	Men: NELS	Men: ELS	Women: NELS	Women: ELS
Reading ($\hat{\beta}$)	0.004*** [0.001]	0.007*** [0.001]	0.010*** [0.002]	0.017*** [0.001]
Change from NELS		0.003* [0.002]		0.007*** [0.003]
IQR	17.260	15.210	15.430	13.440
Reading*IQR	0.064	0.104	0.161	0.231
N	4065	4601	4170	5108

Significance levels: * p < .10, ** p < .05, *** p < .01

The Yitzhaki Decomposition

A standard OLS regression can be decomposed as follows:[4]

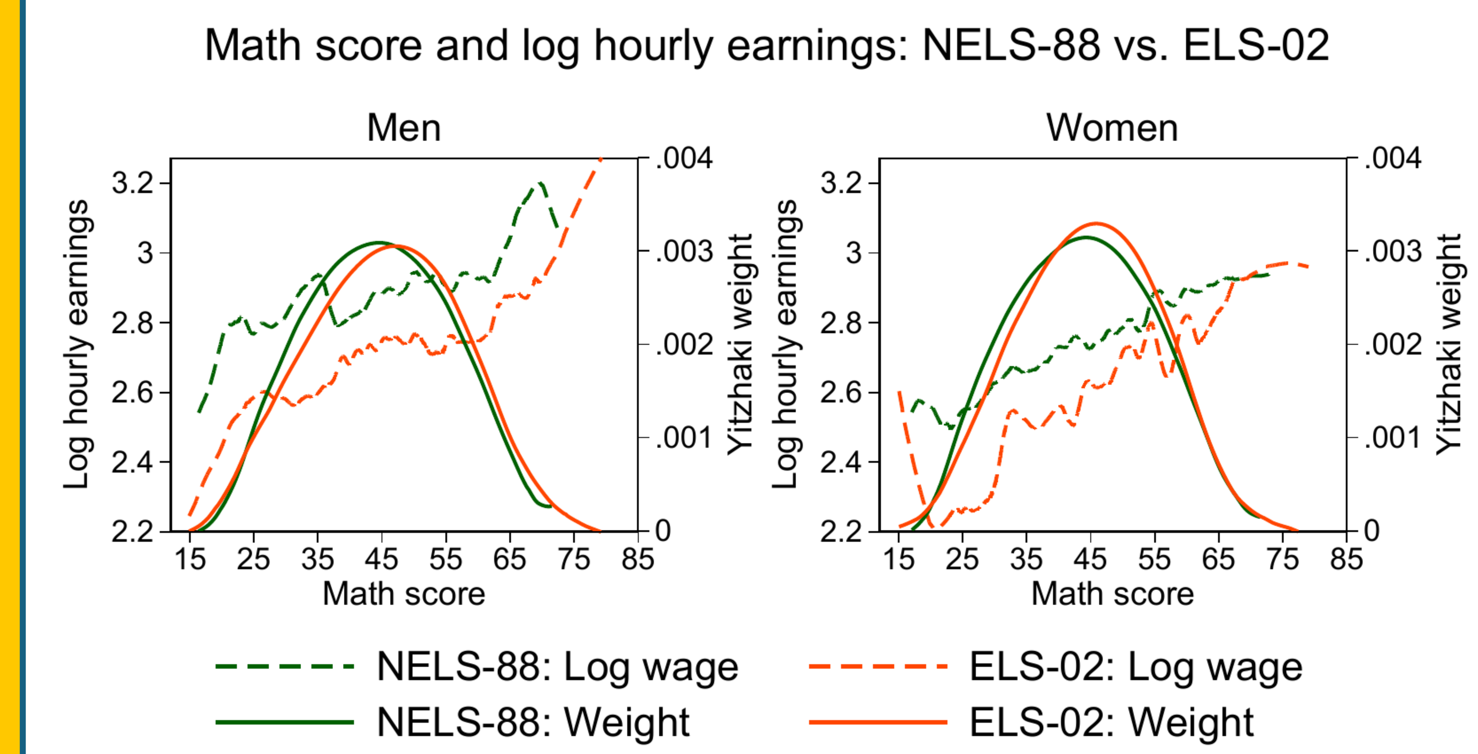
$$\hat{\beta}_{\text{OLS}} = \sum_{j=1}^{n-1} \underset{\text{(Weight)}}{w_j} \times \underset{\text{(Pairwise Slope)}}{s_j} = w_1 s_1 + w_2 s_2 + \cdots + w_{n-1} s_{n-1}$$

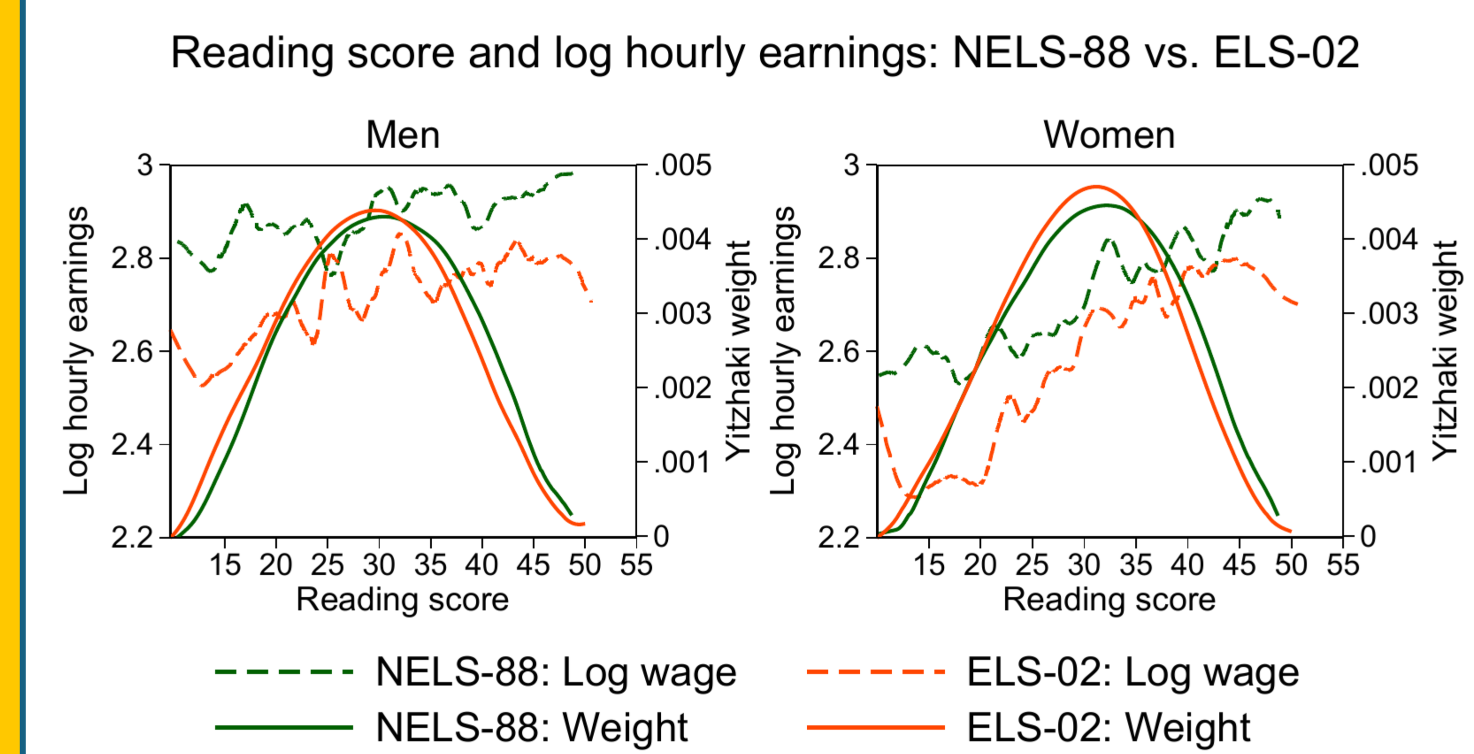
$$\text{with weight properties: } \sum_{j=1}^{n-1} w_j = 1 \text{ and } w_j \geq 0$$

$$\text{calculate pairwise slopes between adjacent score values: } s_j = \frac{y_{j+1} - y_j}{x_{j+1} - x_j}$$

- For both men and women, Yitzhaki weights are similar across cohorts. Yitzhaki weights are calculated only using the score distributions.
- For women, test scores in the lower half of the distribution tend to have higher pairwise slopes and therefore contribute more to the OLS result in NELS, and even more so in ELS.
- For men, test scores across the distribution contribute roughly equally to the OLS result.

Yitzhaki Decomposition Plots





Key Findings

- Between 1990 and 2002, the wage returns to cognitive achievement increased significantly for women. For men, there is less evidence of a significant increase, but no evidence of a decrease.
  - This contrasts with some evidence that wage returns declined for previous cohorts,[5,6] a finding which has been recently challenged by Hellerstein et al.[7]
- The growth in returns to women's test scores appears to be primarily driven by increased wage returns for the bottom half of scorers.

Discussion

- These findings underscore the importance of future research and policy focusing on low-scoring students, whose recent test score declines threaten to widen disparities both in cognitive achievement and in subsequent labor market outcomes.

References: